

Course Level: Undergraduate Open Elective (B. Tech)

BL: 2 Understand

CO: Recall the basics of food groups, food composition, nutritive value and functions of food.

Module 2: Macronutrients: Fats

1. Introduction

Lipids: C, H, and less O; hydrophobic, soluble in organic solvents.

“Lipids” → metabolism; “Fats” → food/diet.

Found in all living cells; complex lipids abundant in brain & nervous tissue.

2. Classification of Lipids

Simple lipids: Esters of fatty acids + alcohols

Fats & oils (mono-, di-, triacylglycerols)

Waxes (FA + long-chain alcohols)

Complex lipids: Yield FA + other groups on hydrolysis

Phospholipids (glycerophospholipids, sphingophospholipids)

Glycolipids, aminolipids, sulpholipids, lipoproteins

Derived lipids: Hydrolysis products retaining lipid properties

3. Key Lipids

Phospholipids: Membrane structure, permeability, fat absorption, lipotropic factors, cholesterol transport.

Cholesterol: Cell membranes, lipoproteins, brain tissue, precursor for bile salts & hormones, insulation for nerve impulses.

4. Fats in the Body

Brown adipose tissue: Heat generation, rich in mitochondria, active in newborns.

Trans fatty acids: Formed by partial hydrogenation; harmful (↑ LDL, ↓ HDL, ↑ CVD risk).

5. Fats in Food

Visible fats: Cooking oils, ghee, butter.

Invisible fats: Integral to cereals, pulses, nuts.

Hidden fats: Processed foods.

Requirements:

Sedentary adults: 25 g/day

Heavy work: 30–40 g/day

Pregnancy/lactation: ~30 g/day

Children/adolescents: 30–50 g/day

Fatty acids:

Saturated (limit <6% calories; meat, dairy, processed foods)

Unsaturated: MUFA (olive, canola), PUFA (sunflower, corn, fish oils)

6. Functions of Fats

- Structural: Cell membranes, organelles.
- Storage: Energy reserve (1 g fat = 9 kcal).
- Metabolic: Precursors for eicosanoids, hormones.
- Absorption: Fat-soluble vitamins (A, D, E, K).
- Insulation & protection: Subcutaneous fat, cushioning organs.
- Regulation: Cell size/shape, blood pressure, reproductive cycle.